

ASHONG RAYMOND NII NOI

01242604B

Assignment 2

Implementing a Queue in Python

```
class Queue:
```

```
def __init__(self):
```

```
# Initializes an empty list to hold queue elements
```

```
self.items = []
```

```
def is_empty(self):
```

```
# Returns True if the queue is empty; otherwise, False
```

```
return len(self.items) == 0
```

```
def enqueue(self, item):
```

```
# Adds an item to the end of the list (end of the queue)
```

```
self.items.append(item)
```

```
def dequeue(self):
```

```
# Removes and returns the first item (front of the queue), if not empty
```

```
return self.items.pop(0) if not self.is_empty() else None
```

```
def size(self):
```

```
# Returns the number of items in the queue
```

```
return len(self.items)
```